

BXP Evidence Index

Claim-to-Evidence Mapping

Every major claim in the BXP evidence package is mapped here to the actual evidence that supports it.

Format

Claim	Evidence	File	Status
Description of claim	What supports it	Where to find it	Honest assessment

Protocol Existence and Design

Claim	Evidence	File	Status
BXP v2.0 protocol specification exists	1,300-line formal specification	SPEC.md	Complete
Protocol was first conceived February 2019	Statement in specification Origin & History	SPEC.md, line 1288	Documented
BXP v1.0 established user-owned data	Listed in CHANGELOG v1.0 section	CHANGELOG.md	Documented
Protocol published to Zenodo with peer review	Zenodo record with DOI 10.5281/zenodo.18906	Published	
Specification covers 31 atmospheric agents	Agent schema section of SPEC.md; Appendix A	SPEC.md §6, Appendix A	Complete
Protocol specifies 5-stage data pipe	SPEC.md §7	SPEC.md §7	Complete
Protocol specifies 20+ REST API endpoints	SPEC.md §8 and reference server	SPEC.md §8, reference-server/server.py	Complete
Agent thresholds aligned with WHO 2019	Threshold values in spec match WHO A	SPEC.md §6, cited in References	Verifiable
Binary file format specified with 32-bit integers	SPEC.md §5.2	SPEC.md §5.2	Specified, not implemented
BXP volume file system architecture	SPEC.md §4	SPEC.md §4	Specified, not physically implemented

Reference Implementation

Claim	Evidence	File	Status
Working reference server exists	~1,877-line FastAPI server	reference-server/server.py	Working
Server accepts POST /bvp/v2/readings	Server code + API docs	reference-server/server.py:581	Implemented
Server uses SQLite for persistence	SQLite database file and database.py	reference-server/database.py	Implemented
Device token authentication implemented	Auth validation in server	reference-server/server.py	Implemented
Rate limiting implemented (per-IP sliding window)	RateLimiter class in server	reference-server/server.py:158	Implemented
Privacy controls implemented (geohash, IP anonymization)	Privacy logic in submit_readings + anonymize_ip	reference-server/server.py	Implemented
Cryptographic deletion with proof of deletion	DELETE endpoint	reference-server/server.py:690	Implemented
Integrity verification endpoint exists	GET /readings/{id}/verify	reference-server/server.py:723	Implemented
Community reports implemented	POST/GET /community/reports	reference-server/server.py	Implemented
Device registration implemented	POST/GET /devices/register	reference-server/server.py	Implemented
Interactive dashboard with map and charts	Dashboard route at /	reference-server/server.py	Implemented
Prometheus metrics endpoint	GET /metrics	reference-server/server.py	Implemented

Embeddable widget		GET /widget/{city}		reference-server/server.py		Implemented
Server can be started with one comma		python server.py	after pip install	-		
reference-server/requirements.txt		Verified				
Implementation published to Zenodo w		DOI 10.5281/zenodo.18907003		https://doi.org/10.5281/zenodo.18907		Published

BXP_HRI (Health Risk Index)

Claim	Evidence	File	Status
BXP_HRI is a composite index incorpo		Formula in SPEC.md §13 and implement	SPEC.md §13, sdk/python/bxp_sdk.py:1 Implemented
Weights derived from WHO DALY burden		Stated in spec with basis; WHO refer	SPEC.md §13.2 Justified; not independently valid
Duration factors (1h/8h/24h) applied		Implemented in SDK and server	bxp_sdk.py, server.py Implemented
Vulnerability factors (general/sensi		Implemented in SDK and server	bxp_sdk.py, server.py Implemented
Six risk levels (CLEAN through HAZAR		Defined in spec and implemented in S	SPEC.md §7 Stage 4, bxp_sdk.py:78 Implemented
BXP_HRI has not been clinically vali		Acknowledged in specification limita	Research report §5.2 ^Δ Limitation acknowledged

Python SDK

Claim	Evidence	File	Status
Python SDK exists		~895-line bxp_sdk.py sdk/python/bxp_sdk.py	Complete
write_bxp() creates valid .bxp.json		Implemented and used in examples	bxp_sdk.py:270 Implemented
read_bxp() verifies integrity on rea		Implemented	bxp_sdk.py:381 Implemented
validate_bxp() checks against spec		Implemented with all required field	bxp_sdk.py:409 Implemented
Synchronous BXPClient implemented		Implemented	bxp_sdk.py Implemented
Asynchronous AsyncBXPClient implemen		Implemented using httpx	bxp_sdk.py Implemented
OfflineQueue with disk persistence i		Implemented	bxp_sdk.py:492 Implemented
pip packaging files present		pyproject.toml and setup.py exist	sdk/python/pyproject.toml, sdk/pytho Present

TypeScript SDK

Claim	Evidence	File	Status
TypeScript SDK exists		bxp-sdk.ts with package.json	sdk/typescript/bxp-sdk.ts Present

CLI Tool

Claim	Evidence	File	Status
CLI tool exists and supports multipl		740+-line bxp_cli.py cli/bxp_cli.py	Complete
generate, read, validate commands im		Source code	cli/bxp_cli.py:129, 193, 254 Implemented
submit and batch-submit implemented		Source code	cli/bxp_cli.py:317, 357 Implemented
export (CSV/GeoJSON) implemented		Source code	cli/bxp_cli.py Implemented
map (HTML generation) implemented		Source code	cli/bxp_cli.py Implemented
BXP_SERVER_URL and BXP_DEVICE_TOKEN		Source code	cli/bxp_cli.py:64, 75 Implemented
Config file (~/.bxp/config.json) imp		Source code	cli/bxp_cli.py:45 Implemented

Sample Dataset

Claim | Evidence | File | Status

Sample dataset contains 10 global ci | JSON file with 10 readings | datasets/sample_readings.bxp.json | Present

Dataset covers 6 continents | Accra, Lagos (Africa), Delhi, Beijin | datasets/sample_readings.bxp.json | Verifiable

All readings use BXP v2.0 schema | bxpVersion: "2.0" in all records | datasets/sample_readings.bxp.json | Verifiable

Ecosystem and Infrastructure

Claim | Evidence | File | Status

MQTT bridge exists | mqtt_bridge.py | integrations/mqtt_bridge.py | Present

Docker deployment configured | Dockerfile and docker-compose.yml | Dockerfile, docker-compose.yml | Present

Postman collection exists | JSON collection file | postman/BXP_Protocol.postman_collect | Present

GitHub Actions CI configured | CI workflow YAML | .github/workflows/ci.yml | Present

CONTRIBUTING.md exists | Contribution guidelines | CONTRIBUTING.md | Present

Claimed But Not Yet Implemented / Verifiable

Claim | Status | Notes

Binary .bxp file format implemented | Not implemented | Specified in SPEC.md §5.2; no encode

Federated node synchronisation | Not implemented | /nodes endpoint exists; P2P sync is

BXP Foundation governance body | Does not exist | Proposed in SPEC.md §12.1

Compatibility with GDPR/CCPA/HIPAA | △ Designed, not audited | Privacy architecture is designed for HL7 FHIR R4 mapping | △ Listed in compatibility matrix; n

External adoption by any organisatio | None yet | Reference implementation is the only

BXP_HRI clinically validated | Not validated | Weights are WHO-derived; no epidemio

Arduino/ESP32 SDKs | Not built | Listed in CHANGELOG §Upcoming

Copyright 2026 Elvarin — Apache 2.0